

Comparison principle for stochastic heat equation on $\mathbb{R}^d$

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Stochastic Heat Equation on $\mathbb{R}^d$

$$\begin{cases} \left(\frac{\partial}{\partial t} - \frac{1}{2} \Delta \right) u(t, x) = \rho(u(t, x)) \dot{W}(t, x), & t > 0, x \in \mathbb{R}^d \\ u(0, \cdot) = \mu(\cdot) \end{cases} \quad (\text{SHE})$$

1. ρ is Lipschitzian.
2. μ is *rough* (measure+integrability condition).
3. $\dot{W}$ is white in time and colored in space (f nonneg. & nonneg. definite):

$$\mathbb{E} \left[\dot{W}(t, x) \dot{W}(s, y) \right] = \delta_0(t - s) f(x - y)$$

► Mild solution

$$u(t, x) = J_0(t, x) + \int_0^t \int_{\mathbb{R}^d} G(t-s, x-y) \rho(u(s, y)) W(ds dy).$$

$$\dagger \quad G(t, x) = (2\pi t)^{-d/2} \exp\left(-\frac{|x|^2}{2t}\right).$$

$$\dagger \quad J_0(t, x) = \int_{\mathbb{R}^d} G(t, x-y) \mu(dy).$$

► Rough initial data: μ is a (nonneg.) measure s.t.

$$J_0(t, x) < \infty \quad \forall t > 0, x \in \mathbb{R}^d \quad \Longleftrightarrow \quad \int_{\mathbb{R}^d} e^{-a|x|^2} \mu(dx) < \infty, \forall a > 0.$$

$$\text{E.g., } \mu = \delta_0, \quad \mu(dx) = e^{-|x|^{3/2}} dx, \quad \mu(dx) = |x|^{-\beta} dx \ (\beta \in (0, d)), \dots$$

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Correlation function f

1. Dalang's condition

$$\int_{\mathbb{R}^d} \frac{\widehat{f}(d\xi)}{1 + |\xi|^2} < \infty. \quad (\text{DL})$$

Necessary and sufficient condition for an $L^2(\Omega)$ -solution. [Dalang' 99]
 $L^2(\Omega)$ continuity.

2. Strengthened Dalang's condition

$$\int_{\mathbb{R}^d} \frac{\widehat{f}(d\xi)}{(1 + |\xi|^2)^{1-\alpha}} < \infty, \quad \text{for some } \alpha \in (0, 1]. \quad (\text{DL}^\alpha)$$

$(\text{DL}) = (\text{DL}^0)$;

Riesz kernel $f(x) = |x|^{-\beta}$, $\beta \in (0, 2 \wedge d)$.

Sample path regularity.

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Rough initial data + (DL) or (DL^α)

- ▶ Weak comparison principle (WCP):

$$\mu_1 \geq \mu_2 \quad \xRightarrow{??} \quad u_1(t, x) \geq u_2(t, x) \quad a.s. \quad \forall t > 0, x \in \mathbb{R}^d.$$

- ▶ Strong comparison principle (SCP):

$$\mu_1 > \mu_2 \quad \xRightarrow{??} \quad u_1(t, x) > u_2(t, x) \quad \forall t > 0, x \in \mathbb{R}^d, \quad a.s.$$

- ▶ Strict positivity:

$$\mu > 0 \quad \xRightarrow{??} \quad u(t, x) > 0 \quad \forall t > 0, x \in \mathbb{R}^d, \quad a.s.$$

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Main gradients

- (1) Workhorse: a moment formula: Exist./unique., moment growth
- (2) Moment increments: Sample path regularity
- (3) Discretization of SHE to a system of infinite-dim SDEs: WCP
- (4) A large deviation result based on (1) and (2).
- (5) Finally, carry out Mueller's arguments using Markov property: SCP

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Brief literature

1. **Mueller '91**: SHE on $\mathbb{R}$ with $\rho(u) = u^\gamma$ with $\gamma \geq 1$.
2. **Shiga '94**: SHE on $\mathbb{R}$ with Lipschitz $\rho(u)$, more general initial data and another way of proving weak comparison principle.
3. **Flores '14**: SHE on $\mathbb{R}$ with delta initial: concentration of measure for discrete directed polymers.
4. **C. & Kim '15**: SHE on $\mathbb{R}$ + fractional Laplacian + Rough initial data.
5. **Gubinelli & Perkowski '15** use paracontrolled distribution to give an intrinsic proof (require initial data strictly positive anywhere).

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6. **Tessitore & Zabczyk '98**: SHE on $\mathbb{R}^d$ with white noise in time colored in space. Function initial data. Very restrictive on f , Riesz kernel case is not included.

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Plan

Moment formula

Sample path regularity

Weak comparison principle

A large deviation result

Strong comparison principle

The workhorse — Moment formula

1. **Theorem.** For any $p \geq 2$, for rough initial data,

$$\text{under (DL)} \quad \|u(t, x)\|_p \leq C(1 + J_0(t, x))H(t; \gamma_p)^{1/2},$$

$$\text{under (DL}^\alpha) \quad \|u(t, x)\|_p \leq C(1 + J_0(t, x)) \exp\left(C \operatorname{Lip}_\rho^{2/\alpha} \rho^{1/\alpha} t\right).$$

2. Remarks:

a. $H(t; \gamma_p)$ is defined through a series, which has at most some exponential growth in t .

b. If initial data is a bounded function, it is immediate that

$$\sup_{(t,x) \in [0,T] \times \mathbb{R}^d} \|u(t, x)\|_p < \infty. \quad (*)$$

Eq. (*) is crucial in the classical theory, which fails for rough initial data.

c. For finite time horizon T , this theorem says that

$$\|u(t, x)\|_p \sim C(1 + J_0(t, x)), \quad \forall (t, x) \in [0, T] \times \mathbb{R}^d.$$

d. $p = 2$ is studied early in [C. & Kim, '15].

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A Gronwall type lemma

► Denote:

$$k(t) := \int_{\mathbb{R}^d} f(z) G(t, z) dz,$$

$$h_n(t) := \int_0^t ds h_{n-1}(s) k(t-s), \quad n \geq 1, \text{ with } h_0(t) := 1,$$

$$H(t; \gamma) := \sum_{n=0}^{\infty} \gamma^n h_n(t), \quad \text{for all } \gamma \geq 0.$$

► **Lemma** For rough initial data, if a nonnegative (measurable) function $g : \mathbb{R}_+ \times \mathbb{R}^d \mapsto \mathbb{R}_+$ satisfies that for all $t > 0$ and $x \in \mathbb{R}^d$,

$$\begin{aligned} g(t, x)^2 \leq 2J_0^2(t, x) + \lambda^2 \int_0^t ds \iint_{\mathbb{R}^{2d}} G(t-s, x-y_1) G(t-s, x-y_2) \\ \times f(y_1 - y_2) g(s, y_1) g(s, y_2) dy_1 dy_2 < \infty, \end{aligned}$$

then

$$g(t, x) \leq J_0(t, x) H(t; 2\lambda^2)^{1/2}.$$

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Bounds for the p -th moments, $p \geq 2$

- For simplicity, suppose $\rho(0) = 0$.

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Plan

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Sample path regularity

Weak comparison principle

A large deviation result

Strong comparison principle

Sample path regularity

1. Recall that the strengthened Dalang's condition

$$\int_{\mathbb{R}^d} \frac{\widehat{f}(d\xi)}{(1 + |\xi|^2)^{1-\alpha}} < \infty, \quad \text{for some } \alpha \in (0, 1]. \quad (\text{DL}^\alpha)$$

2. **Theorem.** Under (DL^α) , for rough initial data, the solution $u(t, x)$ to SHE is (β_1, β_2) -Hölder on $(t, x) \in (0, \infty) \times \mathbb{R}^d$ with

$$\beta_1 \in (0, \alpha/2) \quad \text{and} \quad \beta_2 \in (0, \alpha).$$

3. Remarks:

1. $\alpha = 0$ + logarithm correction terms, $d = 3$, densely blow-up solution, [C., Huang, Khoshnevisan & Kim, '16].
2. $(0, \infty)$ cannot be improved to $[0, \infty)$.
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A flavor of the proof

$$u(t, x) = J_0(t, x) + I(t, x)$$

$$\|I(t, x) - I(t', x')\|_p^2 \leq C I_1(t, x, x') + C I_2(t, t', x') + C I_3(t, t', x'), \quad \text{with}$$

$$\begin{aligned} I_1(t, x, x') &= \int_0^t ds \iint_{\mathbb{R}^{2d}} dy_1 dy_2 f(y_1 - y_2) \\ &\quad \times |G(t - s, x - y_1) - G(t - s, x' - y_1)| \sqrt{1 + \|u(s, y_1)\|_p^2} \\ &\quad \times |G(t - s, x - y_2) - G(t - s, x' - y_2)| \sqrt{1 + \|u(s, y_2)\|_p^2} \end{aligned}$$

$$\begin{aligned} I_2(t, t', x') &= \int_0^t ds \iint_{\mathbb{R}^{2d}} dy_1 dy_2 f(y_1 - y_2) \\ &\quad \times |G(t - s, x' - y_1) - G(t' - s, x' - y_1)| \sqrt{1 + \|u(s, y_1)\|_p^2} \\ &\quad \times |G(t - s, x' - y_2) - G(t' - s, x' - y_2)| \sqrt{1 + \|u(s, y_2)\|_p^2} \end{aligned}$$

$$\begin{aligned} I_3(t, t', x') &= \int_t^{t'} ds \iint_{\mathbb{R}^{2d}} dy_1 dy_2 f(y_1 - y_2) \\ &\quad \times G(t' - s, x' - y_1) \sqrt{1 + \|u(s, y_1)\|_p^2} \\ &\quad \times G(t' - s, x' - y_2) \sqrt{1 + \|u(s, y_2)\|_p^2}. \end{aligned}$$

A flavor of the proof...

$$\|I(t, x) - I(t', x')\|_p^2 \leq Cl_1(t, x, x') + Cl_2(t, t', x') + Cl_3(t, t', x'),$$

with

$$l_1(t, x, x') \leq C|x - x'|^{2\alpha} [J_0(2t, x) + J_0(2t, x')]^2 \int_{\mathbb{R}^d} \frac{\widehat{f}(d\xi)}{(1 + |\xi|^2)^{1-\alpha}}$$

$$l_2(t, t', x') \leq C(t' - t)^\alpha J_0^2(4t', x') \int_{\mathbb{R}^d} \frac{\widehat{f}(d\xi)}{(1 + |\xi|^2)^{1-\alpha}}$$

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1. **Theorem.** Under (DL), let u_1 and u_2 be two solutions to SHE starting from μ_1 and μ_2 , respectively. Then

$$\mu_1 \geq \mu_2 \implies u_1(t, x) \geq u_2(t, x) \text{ a.s., } \forall t \geq 0, x \in \mathbb{R}^d.$$

2. **Corollary.** If, in addition, $\rho(0) = 0$, then

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Ideas in the proof ...

1. Approximate SHE by a system of an infinite dimensional SDEs:

a. A system of infinite particles.

[Mueller '91]

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b. Mollify the unbounded operator Δ .

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Ideas of proof ...

- Use “ $\epsilon > 0$ ” to mollify Δ , $\dot{W}$, μ :

$$\begin{cases} \frac{\partial}{\partial t} u_\epsilon(t, x) = \Delta^\epsilon u_\epsilon(t, x) + \rho(u_\epsilon(t, x)) \dot{W}_x^\epsilon(t), & t > 0, x \in \mathbb{R}^d, \\ u_\epsilon(0, x) = (\mu * G(\epsilon, \cdot))(x), & x \in \mathbb{R}^d, \end{cases}$$

which has a strong solution:

$$u_\epsilon(t, x) = (\mu * G(\epsilon, \cdot))(x) + \int_0^t \Delta^\epsilon u_\epsilon(s, x) ds + \int_0^t \rho(u_\epsilon(s, x)) dW_x^\epsilon(s).$$

- Use standard comparison arguments from SDE to show $u_\epsilon \geq 0$ a.s.
- Lengthy and lengthy computations to show $u_\epsilon \rightarrow u$ in $L^2(\Omega)$ as $\epsilon \downarrow 0$.

Requires a key lemma – Lemma 6.6 of [Shiga 94] (“checked by elementary calculations”). Proved and extended in both [C., Kim '17] and [C., Huang '19].

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A large deviation estimate

1. Fix $\ell, t, M > 0$. Assume $\rho(0) = 0$, $\mu(dx) = 1_{Q(\ell)}(x)dx$, $Q(\ell) := [-\ell, \ell]^d$.

2. **Lemma.** Under (DL^α) , there exist constants $\Theta, m_0 > 0$ s.t. $\forall m \geq m_0$,

$$\begin{aligned} \mathbb{P}\left(u(s, x) \geq \frac{1}{3} 1_{Q(\ell + \frac{M}{m})}(x), \forall (s, x) \in \left[\frac{t}{2m}, \frac{t}{m}\right] \times \mathbb{R}^d\right) \\ \geq 1 - \exp\left(-\Theta m^\alpha (\log m)^{1+\alpha}\right). \end{aligned}$$

A large deviation estimate

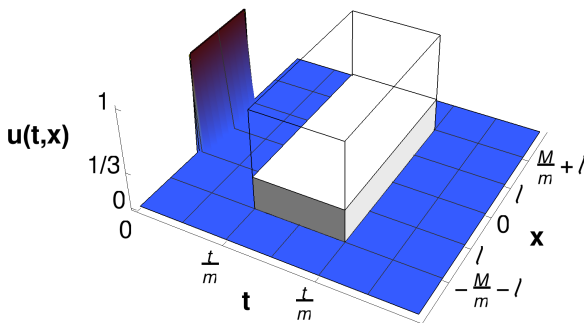
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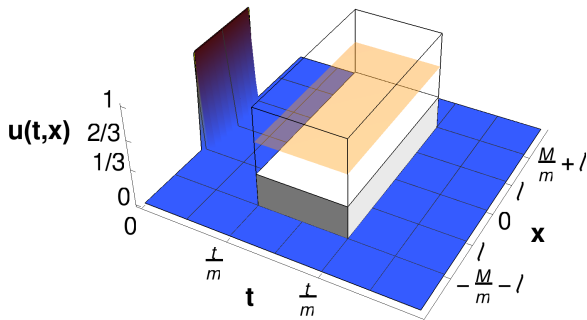
t, M, ℓ given, $\exists m_0$ large enough s.t. $\forall m \geq m_0$

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Ideas of the proof

1. $u(t, x) = J_0(t, x) + I(t, x)$.
2. By choosing m large enough, we have $\inf_{(s,x) \in S} |J_0(t, x)| > 2/3$.
3. The fluctuation of $I(t, x)$ is less than $1/3$ with a high prob.



Some ideas in the proof

- ▶ $u(t, x) = J_0(t, x) + I(t, x)$.

Set $S := [t/(2m), t/m] \times Q(\ell + M/m)$.

It is clear that when m is large enough, $\inf_{(s,x) \in S} |J_0(t, x)| > 2/3$.

- ▶ It reduces to show that with high probability, $\sup_{(s,x) \in S} |I(t, x)| > 1/3$, i.e.,

$$\mathbb{P} \left(\sup_{(s,x) \in S} |I(t, x)| > \frac{1}{3} \right) \leq 3^{-p} \mathbb{E} \left[\sup_{(s,x) \in S} |I(t, x)|^p \right]$$

- ▶ Use the moment increments & Kolmogorov's continuity theorem,

$$\dots \leq C \tau^{\alpha p/4} \exp \left(C p^{1+1/\alpha} \tau \right), \quad \tau := t/m.$$

- ▶ Optimize over p .

This idea appeared earlier in [Conus, Joseph, Khoshnevisan, 12].

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Proof of strong comparison principle

Under (DL^α) : $\mu_1 > \mu_2 \implies u_1(t, x) > u_2(t, x) \quad \forall t > 0, x \in \mathbb{R}^d, \text{ a.s.}$

Setup: Set $v(t, x) = u_1(t, x) - u_2(t, x)$, $\mu = \mu_1 - \mu_2$, $\tilde{\rho}(v) = \rho(v + u_2) = \rho(u_1)$.
Then $\tilde{\rho}(0) = 0$ and $\text{Lip}_\rho = \text{Lip}_{\tilde{\rho}}$. By the weak comparison principle,
 $v(t, x) \geq 0$ for all (t, x) a.s.

Case I Suppose $\mu(dx) = 1_{Q(t)}(x)dx$.

Case II Rough initial data.

Proof of strong comparison principle

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Case II Rough initial data.

Case I-1 Set $\beta = 1/3$ and for $k \geq 0$,

$$B_0 := \left\{ u\left(\frac{t}{2m}, x\right) \geq \beta \mathbb{I}_{S_0^m}(x) \quad \forall x \in \mathbb{R}^d \right\},$$

$$A_k := \left\{ u(s, x) \geq \beta^{k+1} \mathbb{I}_{S_k^m}(x) \quad \forall s \in \left[\frac{(2k+1)t}{2m}, \frac{(k+1)t}{m} \right], \quad \forall x \in \mathbb{R}^d \right\},$$

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with

$$S_k^m := \left(-\ell - \frac{Mk}{m}, \ell + \frac{Mk}{m} \right)^d.$$

Case I-2 By Markov property and the large deviation result,

$$\mathbb{P}(A_k \mid A_{k-1} \cap \cdots \cap A_0) \geq 1 - c(m),$$

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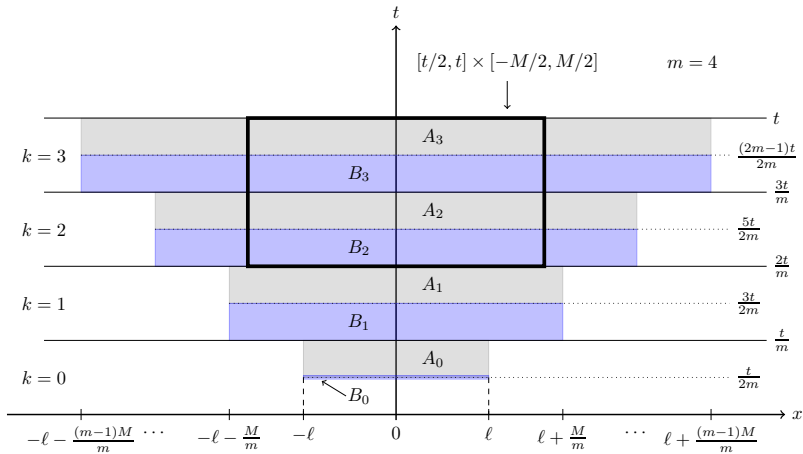
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Case I-3 Notice that

$$\left\{ \inf_{(t,x) \in K} u(t,x) < \beta^m \right\} \subseteq \bigcup_{k=1}^{m-1} (A_k^c \cup B_k^c).$$

Because

$$\begin{aligned} & \mathbb{P}(\cap_{0 \leq k \leq m-1} [A_k \cap B_k]) \\ & \geq 1 - (1 - \mathbb{P}(\cap_{0 \leq k \leq m-1} A_k)) - (1 - \mathbb{P}(\cap_{0 \leq k \leq m-1} B_k)) \\ & \geq (1 - c(m))^{m-1} \mathbb{P}(A_0) + (1 - c(m))^{m-1} \mathbb{P}(B_0) - 1 \\ & \geq 2(1 - c(m))^m - 1, \end{aligned}$$

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Case II For general initial data, fix $\epsilon > 0$. Let $V(t, x) := u(t + \epsilon, x)$. V solves SHE with the time-shifted noise $\dot{W}_\epsilon(t, x) := \dot{W}(t + \epsilon, x)$ starting from $u(\epsilon, x)$ (Markov property).

Claim

$$\mathbb{P} \left(u(\epsilon, x) = 0, \forall x \in \mathbb{R}^d \right) = 0,$$

Prove by contradiction: Using the fact that

$$\lim_{t \rightarrow 0} \int_{\mathbb{R}^d} u(t, x) \phi(x) dx = \int_{\mathbb{R}^d} \phi(x) \mu(dx) \quad \text{in } L^2(\Omega) \text{ for all } \phi \in C_c(\mathbb{R}^d).$$

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